

THEORY AND METHODS
A DIFFERENT VIEW ON EVALUATING THE
PERFORMANCE OF BOOTSTRAP CRITICAL
VALUES

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Summary: Two methods to evaluate the performance of bootstrap-based tests are discussed. The first is one that is traditionally used in the literature, while the second is an alternative method that we propose. To illustrate the differences between the two evaluation methods, we consider testing the mean of a population. Theoretical properties of the corresponding bootstrap critical values are derived. The results of a simulation study are presented to provide empirical comparison of the two techniques with regard to size and power of the test, and to substantiate the theoretical findings. It is found, among others, that the new evaluation method can detect defects of bootstrap critical values that cannot be observed if the current evaluation method is employed. Similar results hold for the “smooth function model”, which allows us to study testing problems where the parameter is for example a variance, a ratio of means or variances, or a correlation coefficient.

1. Introduction

Hall and Wilson (1991) highlighted two important guidelines for bootstrap hypothesis testing. Their first guideline states that, when one wants to

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estimate the critical value, resampling must be done in a way that reflects the null hypothesis. This must be done even if the data were generated from a distribution specified by the alternative hypothesis. This guideline is crucial to the success of bootstrap hypothesis testing and has been mentioned by Beran (1988); Fisher and Hall (1990); Westfall and Young (1993) and recently by Martin (2007). In their second guideline, Hall and Wilson (1991) recommended that bootstrap hypothesis tests should be based on test statistics that are (asymptotically) pivotal. The importance of using pivotal statistics in the bootstrap was considered by Beran (1988) and Hall (1992), among others.

The performance of a newly proposed bootstrap-based test is usually evaluated in order to assess its behaviour with regard to size and power properties. In Section 2 we discuss the evaluation method that is currently used in the literature and then the method we propose for evaluating the performance of a bootstrap-based test. To illustrate the differences between the two evaluation methods, we consider testing the mean of a population in Section 3. Theoretical properties of the corresponding bootstrap critical values are derived and the results of a simulation study are presented to provide empirical comparison of the two techniques. Proofs of theorems are detailed in Appendix A.

2. Evaluation methods

Let $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ be a sample of independent and identically distributed (i.i.d.) random variables with unknown distribution function (d.f.) $F \equiv F_{\boldsymbol{\theta}, \boldsymbol{\sigma}}$, depending on unknown parameter vectors $\boldsymbol{\theta} = \boldsymbol{\theta}(F)$ and $\boldsymbol{\sigma} = \boldsymbol{\sigma}(F)$.

Consider the hypothesis

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \quad \text{vs.} \quad H_A : \boldsymbol{\theta} \in \Theta_A,$$

where Θ_0 and Θ_A are two disjoint subsets of some parameter space $\Theta = \Theta_0 \cup \Theta_A$.

For testing H_0 we consider $\boldsymbol{\sigma}$ as a nuisance parameter, which will be replaced by an estimator that is strongly consistent under both H_0 and H_A . These estimated nuisance parameters will therefore not be indicated in the notation of test statistics and critical values.

Assume, without loss of generality, that the test procedure is of the form:

Reject H_0 if and only if

$$T_n(\mathbf{X}_n) \geq C_n(\alpha; \mathbf{X}_n),$$

where $T_n(\mathbf{X}_n)$ is an appropriate test statistic and $C_n(\alpha; \mathbf{X}_n)$ is a bootstrap critical value, which depends on the significance level α ($0 < \alpha < 1$) of the test and the given set of data $\mathbf{X}_n$. $C_n(\alpha; \mathbf{X}_n)$ is defined as follows:

Let $\mathbf{X}_n^* = (X_1^*, X_2^*, \dots, X_n^*)$ be a bootstrap random sample drawn with replacement from the empirical d.f. (e.d.f.) F_n of $\mathbf{X}_n$. In the bootstrap world (for fixed $\mathbf{X}_n$) we would like to have that $\boldsymbol{\theta}(F_n) \in \Theta_0$. However, this is seldom if ever the case, hence we need to transform the data $X_1, X_2, \dots, X_n$. Denote the transformed variables by

$$W_i^0 = W_i^0(\mathbf{X}_n), \quad i = 1, 2, \dots, n.$$

These variables are chosen such that $\boldsymbol{\theta}(G_n) \in \Theta_0$ in the bootstrap world, where G_n is the e.d.f. of $\mathbf{W}_n^0 = (W_1^0, W_2^0, \dots, W_n^0)$. The bootstrap random sample is given by $\mathbf{W}_n^{0*} = (W_1^{0*}, W_2^{0*}, \dots, W_n^{0*})$ drawn with replacement from G_n . Usually, $\mathbf{W}_n^{0*}$ will depend on both $\mathbf{X}_n$ and $\mathbf{X}_n^*$. The bootstrap estimator $C_n(\alpha; \mathbf{X}_n)$ is now chosen such that

$$P^* (T_n(\mathbf{W}_n^{0*}) \geq C_n(\alpha; \mathbf{X}_n)) \cong \alpha, \quad (1)$$

where P^* denotes the conditional probability law of $\mathbf{W}_n^{0*}$ given $\mathbf{X}_n$. Note that $C_n(\alpha; \mathbf{X}_n)$ can be approximated by a simple Monte-Carlo algorithm suggested by Efron (1979). Throughout the paper this approximation will be denoted by $\hat{C}_n(\alpha; \mathbf{X}_n)$.

Let θ_T denote the true value of the parameter. If $\theta_T \in \Theta_0$ it is denoted by θ^0 , otherwise we write it as θ^A . In order to assess the accuracy of the bootstrap critical value $C_n(\alpha; \mathbf{X}_n)$, the following measure for the size of the test is currently in use in the literature (henceforth referred to as Method I):

$$P_{\theta^0}(T_n(\mathbf{X}_n) \geq C_n(\alpha; \mathbf{X}_n)). \quad (2)$$

See, e.g., Fisher and Hall (1990); Nankervis and Savin (1996); Cao and van Keilegom (2006); Davidson and MacKinnon (2004) and Martin (2007).

The power of the bootstrap-based test is evaluated similarly:

$$P_{\theta^A}(T_n(\mathbf{X}_n) \geq C_n(\alpha; \mathbf{X}_n)). \quad (3)$$

It is important to note that $C_n(\alpha; \mathbf{X}_n)$ is a random variable which depends on the sample. Sakov (1998) remarked that *“this is exactly the problem in using the bootstrap to estimate the critical value. One property that a ‘good’ estimate of the critical value should satisfy is that the critical value estimate should be the same whether the null hypothesis is correct or not. This property does not hold in general when using the bootstrap”*.

Furthermore, it is also evident from (2) and (3) that the data $\mathbf{X}_n$ which are used to derive the bootstrap critical value $C_n(\alpha; \mathbf{X}_n)$ are also used to evaluate its performance. Hence, in general this evaluation technique would typically be too optimistic.

In machine learning literature, however, it is customary to conduct the evaluation of a fitted model (or decision rule), derived from a training sample, by making use of a test sample. The latter is a data set which is representative

of the expected future data points and which is independent of the training sample. It is standard practice in machine learning to extract this test sample from the initial data sample $\mathbf{X}_n$, fit a model on the remaining data (the training sample), and then assess the performance of the model on the test sample. However, as far as the evaluation of bootstrap critical values is concerned, it is unnecessary to split the original data set into two separate data sets, because a test sample can be obtained simply by generating a pseudo sample from the hypothesized null (or alternative) model.

In view of the above remarks, we propose the following alternative evaluation procedure:

Suppose that $\mathbf{V}_n^0 = (V_1^0, V_2^0, \dots, V_n^0)$ is a sequence of i.i.d. pseudo random variables with d.f. $F_{\boldsymbol{\theta}^0, \boldsymbol{\sigma}}$ and assume that $\mathbf{V}_n^0$ is independent of the training sample $\mathbf{X}_n$. In order to assess the accuracy of the bootstrap critical value $C_n(\alpha; \mathbf{X}_n)$, we propose the following measure for the size of the test (henceforth referred to as Method II):

$$P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{X}_n)). \quad (4)$$

For a given realisation $\mathbf{x}_n = (x_1, x_2, \dots, x_n)$ of the data $\mathbf{X}_n$, define

$$\varphi^0(\mathbf{x}_n) = P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{x}_n)),$$

then we have that

$$\begin{aligned} P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{X}_n)) &= E_{\boldsymbol{\theta}} (P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{X}_n) | \mathbf{X}_n)) \\ &= E_{\boldsymbol{\theta}} (\varphi^0(\mathbf{X}_n)), \end{aligned} \quad (5)$$

since, by the independence of $\mathbf{V}_n^0$ and $\mathbf{X}_n$, it follows that

$$\begin{aligned} P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{X}_n) | \mathbf{X}_n = \mathbf{x}_n) &= P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{x}_n) | \mathbf{X}_n = \mathbf{x}_n) \\ &= P(T_n(\mathbf{V}_n^0) \geq C_n(\alpha; \mathbf{x}_n)) \\ &= \varphi^0(\mathbf{x}_n). \end{aligned}$$

The V_n^0 pseudo random vectors are used to determine *typical values* of the test statistic under H_0 . These values are then used to evaluate the performance of the (fixed) bootstrap critical value $C_n(\alpha; \mathbf{x}_n)$.

The power of the bootstrap-based test can be evaluated similarly:

$$P(T_n(\mathbf{V}_n^A) \geq C_n(\alpha; \mathbf{X}_n)) = E_{\boldsymbol{\theta}}(\varphi^A(\mathbf{X}_n)), \quad (6)$$

where $\mathbf{V}_n^A = (V_1^A, V_2^A, \dots, V_n^A)$ is a sequence of i.i.d. pseudo random variables with d.f. $F_{\boldsymbol{\theta}^A, \boldsymbol{\sigma}}$, which is independent of the training sample $\mathbf{X}_n$, and

$$\varphi^A(\mathbf{x}_n) = P(T_n(\mathbf{V}_n^A) \geq C_n(\alpha; \mathbf{x}_n)).$$

From (5) and (6) it is clear that choosing $\boldsymbol{\theta}$ -values which differ slightly from $\boldsymbol{\theta}^0$ (or $\boldsymbol{\theta}^A$) will enable one to determine the robustness of $C_n(\alpha; \mathbf{X}_n)$ as far as size (or power) is concerned.

The alternative evaluation method was mentioned in Allison and Swanepoel (2009). However, no theoretical or empirical justification was given there. We shall now contribute to this by applying evaluation Methods I and II in an illustrative example by considering testing the mean of a population.

It is important to keep in mind that we are concerned with proposing a new method of evaluating the performance of bootstrap critical values and not with a new bootstrap procedure.

3. The mean in the univariate case

Let $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ denote a random sample from an unknown univariate d.f. F with finite mean μ and unknown variance (nuisance parameter) σ^2 , $0 < \sigma^2 < \infty$. In this section we shall present some theoretical properties of the bootstrap estimate of the critical value for the test

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_A : \mu > \mu_0,$$

based on evaluation Methods I and II. This will be done using the “right” and “wrong” bootstrap critical values for the pivotal and non-pivotal test statistics. The term “right” refers to the case where resampling is done from the transformed data $\mathbf{W}_n^0$ in order to “mimic” H_0 in the bootstrap world, whereas the term “wrong” refers to resampling being done from the original sample $\mathbf{X}_n$. In what follows, the subscripts P and NP will refer to the pivotal and non-pivotal cases, respectively, whereas the superscripts W and R will refer to the wrong and right critical values, respectively.

Throughout the discussion below we shall assume that

$$X_1 - \mu \stackrel{d}{=} V_1^0 - \mu_0,$$

where $\mu_0 = E(V_1^0)$.

Let $\mu_\nu = E((X_1 - \mu)^\nu)$, $\nu = 2, 3, \dots$, $\kappa_3 = \mu_3/\sigma^3$, $\kappa_4 = \mu_4/\sigma^4 - 3$ and $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$, $0 < \alpha < 1$, with Φ the standard normal d.f., with corresponding density function ϕ . Further, denote $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, $S_n^2(\mathbf{X}_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$, $\bar{X}_n^* = \frac{1}{n} \sum_{i=1}^n X_i^*$, $S_n^2(\mathbf{X}_n^*) = \frac{1}{n} \sum_{i=1}^n (X_i^* - \bar{X}_n^*)^2$, $\bar{V}_n^0 = \frac{1}{n} \sum_{i=1}^n V_i^0$ and $S_n^2(\mathbf{V}_n^0) = \frac{1}{n} \sum_{i=1}^n (V_i^0 - \bar{V}_n^0)^2$.

We shall consider four scenarios. For each of the scenarios, theorems relating to the two different evaluation methods are stated that describe the behaviour of the bootstrap critical value with respect to size properties.

Scenario 1: Using a non-pivotal test statistic and the wrong critical value

The test rejects H_0 if and only if

$$T_{n,NP}(\mathbf{X}_n) = \sqrt{n}(\bar{X}_n - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n),$$

where $C_{n,NP}^W(\alpha; \mathbf{X}_n)$ is defined by

$$P^* \left(\sqrt{n}(\bar{X}_n^* - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n) \right) \cong \alpha.$$

Denote this test procedure by (W,NP).

Theorem 1 (Method I). If $E(|X_1|^3) < \infty$ then, as $n \rightarrow \infty$,

$$P_{H_0}(\sqrt{n}(\bar{X}_n - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n)) \rightarrow \begin{cases} 1 & , \text{ when } \alpha \geq 0.5 \\ 0 & , \text{ when } \alpha < 0.5. \end{cases}$$

Theorem 2 (Method II). Suppose that $E(|X_1|^3) < \infty$. Then, as $n \rightarrow \infty$,

$$P\left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n)\right) \rightarrow \begin{cases} 1 - \Phi\left(\frac{z_{1-\alpha}}{\sqrt{2}}\right) & , \text{ when } \mu = \mu_0 \\ 1 & , \text{ when } \mu < \mu_0 \\ 0 & , \text{ when } \mu > \mu_0. \end{cases}$$

Scenario 2: Using a pivotal test statistic and the wrong critical value

The test rejects H_0 if and only if

$$T_{n,P}(\mathbf{X}_n) = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n),$$

where $C_{n,P}^W(\alpha; \mathbf{X}_n)$ is defined by

$$P^*\left(\frac{\sqrt{n}(\bar{X}_n^* - \mu_0)}{S_n(\mathbf{X}_n^*)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n)\right) \cong \alpha.$$

Denote this test procedure by (W,P).

Theorem 3 (Method I). If $E(X_1^4) < \infty$ then, as $n \rightarrow \infty$,

$$P_{H_0}\left(\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n)\right) \rightarrow \begin{cases} 1 & , \text{ when } \alpha \geq 0.5 \\ 0 & , \text{ when } \alpha < 0.5. \end{cases}$$

Theorem 4 (Method II). Suppose that $E(X_1^4) < \infty$. Then, as $n \rightarrow \infty$,

$$P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n)\right) \rightarrow \begin{cases} 1 - \Phi\left(\frac{z_{1-\alpha}}{\sqrt{2}}\right) & , \text{ when } \mu = \mu_0 \\ 1 & , \text{ when } \mu < \mu_0 \\ 0 & , \text{ when } \mu > \mu_0. \end{cases}$$

Scenario 3: Using a non-pivotal test statistic and the right critical value

The test rejects H_0 if and only if

$$T_{n,NP}(\mathbf{X}_n) = \sqrt{n}(\bar{X}_n - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n),$$

where $C_{n,NP}^R(\alpha; \mathbf{X}_n)$ is defined by

$$P^* \left(\sqrt{n}(\bar{W}_n^{0*} - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha, \quad (7)$$

$\bar{W}_n^{0*} = \frac{1}{n} \sum_{i=1}^n W_i^{0*}$ and $W_i^{0*} = X_i^* - \bar{X}_n + \mu_0$, $i = 1, 2, \dots, n$. Hence, (7) reduces to

$$P^* \left(\sqrt{n}(\bar{X}_n^* - \bar{X}_n) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha.$$

Denote this test procedure by (R,NP).

Theorem 5 (Method I). Suppose $E(|X_1|^5) < \infty$ and that Cramér's continuity condition holds, i.e.,

$$\limsup_{|t| \rightarrow \infty} |\chi(t)| < 1,$$

where $\chi(t)$ is the characteristic function of X_1 , then as $n \rightarrow \infty$,

$$P_{H_0}(\sqrt{n}(\bar{X}_n - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n)) = \alpha + C_n^I + O(n^{-1}),$$

where

$$C_n^I = -\frac{\kappa_3 z_{1-\alpha}^2 \phi(z_{1-\alpha})}{2n^{1/2}}.$$

Theorem 6 (Method II). Suppose $E(|X_1|^5) < \infty$ and that Cramér's continuity condition holds. Then, as $n \rightarrow \infty$,

$$P \left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n) \right) = \alpha + C_n^{II} + O(n^{-3/2}),$$

where

$$C_n^{II} = \frac{z_{1-\alpha} \phi(z_{1-\alpha}) \{z_{1-\alpha}^2 (\mu_4 - \sigma^4) + (\mu_4 + 3\sigma^4)\}}{8\sigma^4 n}.$$

Scenario 4: Using a pivotal test statistic and the right critical value

The test rejects H_0 if and only if

$$T_{n,P}(\mathbf{X}_n) = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n),$$

where $C_{n,P}^R(\alpha; \mathbf{X}_n)$ is defined (making use of W_i^{0*} , $i = 1, 2, \dots, n$, again) by

$$P^* \left(\frac{\sqrt{n}(\bar{X}_n^* - \bar{X}_n)}{S_n(\mathbf{X}_n^*)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha.$$

Denote this test procedure by (R,P).

Theorem 7 (Method I). Suppose $E(X_1^6) < \infty$ and that Cramér's continuity condition holds, then as $n \rightarrow \infty$,

$$P_{H_0} \left(\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) = \alpha + D_n^I + O(n^{-3/2}),$$

where

$$D_n^I = \frac{z_{1-\alpha}(1 + 2z_{1-\alpha}^2)\phi(z_{1-\alpha})(\kappa_4 - \frac{3}{2}\kappa_3^2)}{6n}.$$

Theorem 8 (Method II). Suppose $E(X_1^6) < \infty$ and that Cramér's continuity condition holds. Then, as $n \rightarrow \infty$, we have that

$$P \left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) = \alpha + D_n^{II} + O(n^{-2}),$$

where

$$D_n^{II} = \frac{(1 + 2z_{1-\alpha}^2)\phi(z_{1-\alpha})\{\mu_3(21\sigma^4 + 15\mu_4) - 12\sigma^2\mu_5\}}{48\sigma^7 n^{3/2}}.$$

Remarks:

- (a) If $\mu = \mu_0$, the probability of a Type I-error derived in Theorems 1 and 3 (Method I) converges to a discontinuous step function of α . This is in contrast with the continuous limit function of α obtained in Theorems 2 and 4 (Method II). The latter seems to be a more realistic limit behaviour of the probability of a Type I-error.
- (b) From Theorem 5 it follows that if $\kappa_3 \geq 0$ (positive skewed distribution), then $C_n^I \leq 0$. This property will further be established in the Monte-Carlo simulation below for the exponential and chi-squared distributions for small to moderate sample sizes.
- (c) Note that $C_n^{II} \geq 0$ in Theorem 6 for $\alpha \leq 0.5$. Hence, (R,NP) is a liberal test for large samples. The Monte-Carlo study below will

also show that this property holds for small to moderate sample sizes irrespective of the distributions considered there.

- (d) For the distributions considered in the Monte-Carlo study below, D_n^I given in Theorem 7 is negative for the uniform and chi-squared distributions, zero for the normal and exponential distributions and positive for the double exponential distribution. This behaviour of D_n^I is also reflected in the simulation study for small to moderate sample sizes.
- (e) The constant D_n^{II} is zero for all symmetric distributions.
- (f) It is evident from Theorems 5–8 that the probability of a Type I-error incurred when using evaluation Method II, converges to α faster than when using evaluation Method I.
- (g) For the proofs of Theorems 1, 3, 5 and 7 the reader is referred to Hall (1992).
- (h) The proofs of Theorems 2, 4, 6 and 8 are deferred to Appendix A.

4. Monte-Carlo study

In this section we present the results of a Monte-Carlo study, based on evaluation Methods I and II, in order to compare the sizes and powers of (R,P) and (R,NP) for sample sizes $n = 20, 30, 50, 100$ and 200 , respectively. Since we found that the Monte-Carlo results for (W,P) and (W,NP) were in agreement with the theoretical results derived in Theorems 1–4, we will not present them here. Subscripts I and II will be used in the notation of these test procedures to distinguish between the two evaluation methods. A nominal significance level of $\alpha = 0.05$ was used. Bootstrap critical values were based on $B = 1\,000$ independent replications. All calculations were done using double precision arithmetic in FORTRAN and routines from the IMSL library. For size estimates the $\mathbf{X}_n$ -data and the pseudo random variables $\mathbf{V}_n^0$ were generated from five different distributions with mean μ_0 :

- Normal distribution with variance 1;
- Uniform distribution with variance 1/12;
- Double exponential distribution with variance 2;
- Exponential distribution with variance 1;
- Chi-squared distribution with variance 6.

A limited power study was also conducted, where the $\mathbf{X}_n$ -data and the pseudo random variables $\mathbf{V}_n^A$ were generated from the following two distributions with mean μ_A :

- Double exponential distribution with variance 2;
- Exponential distribution with variance 1.

For the evaluation based on Method I, size (or power) estimates were calculated as the proportion of 20 000 Monte-Carlo samples that resulted in rejection of H_0 (or H_A) using the bootstrap critical values $\hat{C}_{n,NP}^R(\alpha; \mathbf{X}_n)$ and $\hat{C}_{n,P}^R(\alpha; \mathbf{X}_n)$, respectively. The standard errors of the estimated probabilities are less than or equal to $\sqrt{0.25/20\,000} = 0.0035$.

For the evaluation based on Method II, size estimates ($\bar{p}$) were calculated according to

$$\bar{p} = \frac{1}{MC_1} \sum_{j=1}^{MC_1} p_j,$$

where

$$p_j = \frac{1}{MC_2} \sum_{i=1}^{MC_2} I\left(T_n(\mathbf{V}_{n,i}^0) \geq \hat{C}_n(\alpha; \mathbf{X}_n^j)\right), \quad j = 1, 2, \dots, MC_1.$$

The statistic T_n was replaced by $T_{n,P}$ or $T_{n,NP}$, while $\hat{C}_n(\alpha; \mathbf{X}_n^j)$ was replaced by $\hat{C}_{n,P}^R(\alpha; \mathbf{X}_n^j)$ or $\hat{C}_{n,NP}^R(\alpha; \mathbf{X}_n^j)$ for independent samples $\mathbf{X}_n^j$, $j = 1, 2, \dots, MC_1$.

The standard errors of the estimated sizes were calculated as

$$\sqrt{\frac{\frac{1}{MC_1} \sum_{j=1}^{MC_1} (p_j - \bar{p})^2}{MC_1}}.$$

Power estimates were calculated similarly by replacing $\mathbf{V}_{n,i}^0$ by $\mathbf{V}_{n,i}^A$. In the Monte-Carlo study we took $MC_1 = 15\,000$ and $MC_2 = 10\,000$. All standard errors were found to be negligibly small and are not reported in the tables.

Table 1 Estimated sizes using the “right” critical values

Distribution	μ_0	n	$(R,P)_I$	$(R,P)_{II}$	$(R,NP)_I$	$(R,NP)_{II}$
Normal	0.0	20	0.048	0.052	0.060	0.064
		30	0.049	0.052	0.059	0.059
		50	0.049	0.051	0.055	0.056
		100	0.051	0.050	0.054	0.053
		200	0.050	0.050	0.050	0.052
Uniform	0.5	20	0.037	0.054	0.065	0.060
		30	0.043	0.053	0.059	0.057
		50	0.048	0.051	0.058	0.054
		100	0.049	0.051	0.054	0.052
		200	0.050	0.050	0.052	0.052
Double exponential	0.0	20	0.062	0.053	0.060	0.072
		30	0.060	0.051	0.056	0.066
		50	0.057	0.051	0.053	0.060
		100	0.054	0.050	0.051	0.055
		200	0.050	0.050	0.050	0.053
Exponential	1.0	20	0.038	0.043	0.024	0.080
		30	0.044	0.046	0.025	0.072
		50	0.047	0.047	0.028	0.064
		100	0.050	0.049	0.031	0.058
		200	0.050	0.050	0.036	0.054
Chi-squared	3.0	20	0.043	0.045	0.030	0.076
		30	0.045	0.046	0.030	0.068
		50	0.048	0.048	0.032	0.061
		100	0.050	0.050	0.033	0.056
		200	0.050	0.050	0.039	0.054

Table 2 Estimated powers using the “right” critical values

Distribution	μ_A	n	(R,P) _I	(R,P) _{II}	(R,NP) _I	(R,NP) _{II}
Double Exponential	0.2	20	0.188	0.161	0.189	0.192
		30	0.218	0.197	0.222	0.221
		50	0.281	0.265	0.285	0.279
		100	0.411	0.413	0.419	0.419
		200	0.640	0.640	0.645	0.644
	0.4	20	0.376	0.356	0.411	0.404
		30	0.476	0.464	0.506	0.489
		50	0.621	0.637	0.651	0.656
		100	0.874	0.876	0.889	0.883
		200	0.981	0.989	0.993	0.990
Exponential	1.2	20	0.244	0.238	0.164	0.248
		30	0.322	0.321	0.223	0.298
		50	0.462	0.467	0.351	0.398
		100	0.718	0.719	0.637	0.616
		200	0.931	0.930	0.904	0.874
	1.4	20	0.699	0.678	0.562	0.545
		30	0.853	0.840	0.753	0.680
		50	0.969	0.964	0.939	0.869
		100	0.998	0.999	0.998	0.991
		200	1.000	1.000	1.000	0.995

From Tables 1 and 2 we conclude the following:

- (i) The estimated sizes of (R,P) and (R,NP) converge to the nominal significance level as n increases, for both evaluation methods. This agrees with the results of Theorems 5–8.
- (ii) The estimated sizes of (R,P) based on Method II attain the nominal significance level for symmetric distributions, even for *small* values of n . This is consistent with the result derived in Theorem 8, in the sense that the constant D_n^{II} is 0 for symmetric distributions and the rest term is $O(n^{-2})$. However, the estimated sizes of (R,P) based on Method I is slightly conservative for the uniform distribution and liberal for the double exponential distributions.
- (iii) For the asymmetric distributions the estimated sizes of (R,P) are more or

less the same for the two evaluation methods.

- (iv) The estimated sizes of (R,NP) based on Method II decreases monotonically to the nominal significance level as n increases. This is in accordance with the result of Theorem 6, where it was shown that the constant $C_n^{II} \geq 0$. This test, therefore, tends to be liberal. However, this is not the case if (R,NP) is evaluated using Method I: for asymmetric distributions (R,NP) tends to be conservative (which agrees with the result in Theorem 5), while for the symmetric distributions the sizes of (R,NP) are closer to the nominal significance level than those obtained when using Method II.
- (v) Based on Method II, the estimated sizes of (R,P) converge quicker to the nominal significance level than those of (R,NP). This agrees with the results of Theorems 6 and 8, where it was shown that

$$P\left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n)\right) = \alpha + O(n^{-1}),$$

and

$$P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n)\right) = \alpha + O(n^{-3/2}).$$

- (vi) Method II clearly shows that (R,P) always outperforms (R,NP) in terms of estimated sizes, which is consistent with the second guideline given by Hall and Wilson (1991). However, this is not the case if the evaluation is based on Method I (see, e.g., the double exponential distribution).
- (vii) For the exponential distribution the estimated sizes of (R,NP), when using evaluation Method I, are well below the specified nominal significance level, whereas the estimated sizes of (R,NP), based on Method II, are greater than this nominal level. However, when one considers the powers given in Table 2, one finds that as one moves further away from the null hypothesis, the powers of (R,NP) based on Method I begin to dominate the powers of (R,NP) based on Method II.
- (viii) When one considers the double exponential distribution, one finds that the behaviour of the estimated powers of (R,NP) are similar to those found for the exponential distribution described in (vii). However, here the estimated sizes of (R,NP) based on both evaluation methods are greater than the nominal significance level, but still have the property that the estimated size based on Method II is greater than the estimated size based on Method I.

- (ix) It therefore seems, in the light of (iv) and (vi) – (viii) above, that evaluation Method I is too optimistic in the evaluation of (R, NP) .

5. Concluding remarks

It is clear from the discussions in this paper, that it is preferable to make use of a pivotal test statistic and the “right” bootstrap critical value. These findings are in agreement with the two guidelines proposed by Hall and Wilson (1991). It is further recommended that bootstrap critical values should be evaluated using the new evaluation method as it can detect defects of bootstrap estimated critical values which cannot be observed if one uses the current evaluation method.

In this paper we considered testing the mean of a population so that we could present our findings in a comprehensive manner. However, similar results hold for the smooth function model, which allows us to study problems where the parameter is a variance, a ratio of means or variances, a difference of means or variances, a correlation coefficient, etc. The proofs of our theorems rely heavily on using valid Edgeworth and Cornish-Fisher expansions for the sample mean. Hall (1992) studied the smooth function model and established expansions for this model rigorously, both in the real world and the bootstrap world. Hence, by using these expansions, results analogous to those in Theorems 1–8 can be derived in this more general setting.

Appendix A: Proofs of Theorems

A.1 Proof of Theorem 2

By applying the Berry-Esséen theorem in the bootstrap world (with reference to P^*) we deduce that

$$\begin{aligned} \alpha &\cong P^* \left(\sqrt{n}(\bar{X}_n^* - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n) \right) \\ &= 1 - P^* \left(\frac{\sqrt{n}(\bar{X}_n^* - \bar{X}_n)}{S_n(\mathbf{X}_n)} \leq \frac{C_{n,NP}^W(\alpha; \mathbf{X}_n)}{S_n(\mathbf{X}_n)} - \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \right) \\ &= 1 - \Phi \left(\frac{C_{n,NP}^W(\alpha; \mathbf{X}_n)}{S_n(\mathbf{X}_n)} - \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \right) + O_{a.s.}(n^{-1/2}). \end{aligned}$$

Hence, as $n \rightarrow \infty$, we have that

$$C_{n,N-P}^W(\alpha; \mathbf{X}_n) = \sqrt{n}(\bar{X}_n - \mu_0) + S_n(\mathbf{X}_n)z_{1-\alpha} + S_n(\mathbf{X}_n)o_{a.s.}(1). \quad (8)$$

Using the Berry-Esséen theorem in the real world (with reference to P) we find that

$$\begin{aligned} \varphi^0(\mathbf{X}_n) &\equiv P \left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n) | \mathbf{X}_n \right) \\ &= 1 - \Phi \left(\frac{C_{n,NP}^W(\alpha; \mathbf{X}_n)}{\sigma} \right) + O(n^{-1/2}), \end{aligned}$$

where $O(n^{-1/2})$ is a non-random rest term.

From (8), the CLT and Kolmogorov's Strong Law of Large Numbers it follows that if $\mu = \mu_0$,

$$\varphi^0(\mathbf{X}_n) \xrightarrow{d} 1 - \Phi(Z + z_{1-\alpha}) \quad \text{as } n \rightarrow \infty, \quad (9)$$

where Z is a $N(0, 1)$ random variable.

Consequently, we conclude from (9), the boundedness of φ^0 and Theorem 5.4 of Billingsley (1968) that

$$\begin{aligned} P\left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^W(\alpha; \mathbf{X}_n)\right) &= E_\mu(\varphi^0(\mathbf{X}_n)) \\ &\rightarrow 1 - E(\Phi(Z + z_{1-\alpha})) \\ &= 1 - \Phi\left(\frac{z_{1-\alpha}}{\sqrt{2}}\right). \end{aligned} \quad (10)$$

The last step follows from the fact that

$$E(\Phi(Z + z_{1-\alpha})) = \Phi \star \Phi(z_{1-\alpha}),$$

where $\star$ denotes convolution.

Clearly, from (8) and Kolmogorov's Strong Law of Large Numbers we conclude that the probability on the left-hand side of (10) converges to 1 or 0 if $\mu < \mu_0$ or $\mu > \mu_0$, respectively.

A.2 Proof of Theorem 4

Clearly,

$$\begin{aligned} &P^*\left(\frac{\sqrt{n}(\bar{X}_n^* - \mu_0)}{S_n(\mathbf{X}_n^*)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n)\right) \cong \alpha \\ \Leftrightarrow &P^*\left(\frac{\sqrt{n}(\bar{X}_n^* - \bar{X}_n)}{S_n(\mathbf{X}_n^*)} + \frac{\Delta_n^*}{\sqrt{n}} \leq C_{n,P}^W(\alpha; \mathbf{X}_n) - \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)}\right) \\ \cong &1 - \alpha, \end{aligned} \quad (11)$$

where

$$\Delta_n^* = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \cdot \frac{\sqrt{n}(S_n(\mathbf{X}_n) - S_n(\mathbf{X}_n^*))}{S_n(\mathbf{X}_n^*)} = \Delta_{n1} \cdot \Delta_{n2}^*.$$

The CLT implies that, under H_0 , $\Delta_{n1} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$, so that $\Delta_{n1} = O_P(1)$. Also, applying classical bootstrap theory it readily follows, using the assumption $E(X_1^4) < \infty$, that

$$\Delta_{n2}^* \xrightarrow{d^*} N(0, (\mu_4 - \sigma^4)/4\sigma^4) \quad \text{a.s. as } n \rightarrow \infty, \quad (12)$$

where d^* denotes convergence in conditional, given $\mathbf{X}_n$, distribution (i.e. weak convergence in the bootstrap world).

Let $a_n = n^{-1/2} \log n$. Then, using the fact that $\Delta_{n1} = O_P(1)$ and (12) we have that

$$\begin{aligned} P^*(n^{-1/2}|\Delta_n^*| > a_n) &= P^*(|\Delta_{n1}| \cdot |\Delta_{n2}^*| > \log n) \\ &= o_{a.s.}(1) \quad \text{as } n \rightarrow \infty. \end{aligned} \quad (13)$$

From (11), (13) and the Lemma given on pp. 228 – 229 by Serfling (1980) we therefore have that

$$\Phi \left(C_{n,P}^W(\alpha; \mathbf{X}_n) - \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} \right) = 1 - \alpha + o_{a.s.}(1),$$

implying that

$$C_{n,P}^W(\alpha; \mathbf{X}_n) = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S_n(\mathbf{X}_n)} + z_{1-\alpha} + o_{a.s.}(1). \quad (14)$$

Using the Berry-Esséen theorem for a Student t-statistic (see, e.g., Bentkus and Götze, 1996) we find that

$$\begin{aligned} \varphi^0(\mathbf{X}_n) &\equiv P \left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n) \middle| \mathbf{X}_n \right) \\ &= 1 - \Phi(C_{n,P}^W(\alpha; \mathbf{X}_n)) + O(n^{-1/2}), \end{aligned}$$

where $O(n^{-1/2})$ is a non-random rest term.

From (14), the CLT, Kolmogorov's Strong Law of Large Numbers and Slutsky's theorem, it follows that if $\mu = \mu_0$,

$$\varphi^0(\mathbf{X}_n) \xrightarrow{d} 1 - \Phi(Z + z_{1-\alpha}) \quad \text{as } n \rightarrow \infty, \quad (15)$$

where Z is a $N(0, 1)$ random variable.

Hence, from (15), the boundedness of φ^0 and Theorem 5.4 of Billingsley (1968) it follows that

$$\begin{aligned}
P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^W(\alpha; \mathbf{X}_n)\right) &= E_\mu(\varphi^0(\mathbf{X}_n)) \\
&\rightarrow 1 - E(\Phi(Z + z_{1-\alpha})) \\
&= 1 - \Phi\left(\frac{z_{1-\alpha}}{\sqrt{2}}\right). \quad (16)
\end{aligned}$$

Furthermore, it is clear from (14) and Kolmogorov's Strong Law of Large Numbers that the left-hand side of (16) converges to 1 or 0 if $\mu < \mu_0$ or $\mu > \mu_0$, respectively.

A.3 Proof of Theorem 6

To prove Theorem 6, we shall make use of the following Lemma.

Lemma 1. Suppose that $E(|X_1|^5) < \infty$. Then, as $n \rightarrow \infty$,

$$\begin{aligned}
\text{(a)} \quad E(S_n(\mathbf{X}_n) - \sigma) &= -\frac{(\mu_4 + 3\sigma^4)}{8n\sigma^3} + O(n^{-2}), \\
\text{(b)} \quad E((S_n(\mathbf{X}_n) - \sigma)^2) &= \frac{\mu_4 - \sigma^4}{4n\sigma^2} + O(n^{-2}).
\end{aligned}$$

Proof. Using a Taylor series expansion we deduce that

$$E(S_n(\mathbf{X}_n)) = (E(S_n^2(\mathbf{X}_n)))^{1/2} - \frac{1}{8} \cdot \frac{\text{Var}(S_n^2(\mathbf{X}_n))}{(E(S_n^2(\mathbf{X}_n)))^{3/2}} + O(n^{-2}).$$

Applying the result in (27.4.1) and (27.4.2) of Cramér (1945) we conclude that

$$\begin{aligned}
E(S_n(\mathbf{X}_n)) &= \sigma \left(1 - \frac{1}{n}\right)^{1/2} - \frac{1}{8\sigma^3} \left(1 - \frac{1}{n}\right)^{-3/2} \frac{(\mu_4 - \sigma^4)}{n} + O(n^{-2}) \\
&= \sigma - \frac{\sigma}{2n} - \frac{1}{8\sigma^3} \frac{(\mu_4 - \sigma^4)}{n} + O(n^{-2}) \\
&= \sigma - \frac{(\mu_4 + 3\sigma^4)}{8n\sigma^3} + O(n^{-2}),
\end{aligned}$$

which proves (a).

Furthermore, from (a) and (27.7.2) of Cramér (1945) it follows that

$$\begin{aligned}
E((S_n(\mathbf{X}_n) - \sigma)^2) &= \text{Var}(S_n(\mathbf{X}_n)) + \{E(S_n(\mathbf{X}_n)) - \sigma\}^2 \\
&= \frac{\mu_4 - \sigma^4}{4n\sigma^2} + O(n^{-2}),
\end{aligned}$$

which proves (b).

We now proceed with the proof of the theorem:

From Hall (1992) it follows that a Cornish-Fisher expansion for $C_{n,NP}^R(\alpha; \mathbf{X}_n)$ is given by

$$\begin{aligned}
C_{n,NP}^R(\alpha; \mathbf{X}_n) &= S_n(\mathbf{X}_n) \left\{ z_{1-\alpha} + n^{-1/2} \hat{P}_1^{cf}(z_{1-\alpha}) \right. \\
&\quad \left. + n^{-1} \hat{P}_2^{cf}(z_{1-\alpha}) + O_{a.s.}(n^{-3/2}) \right\}, \quad (17)
\end{aligned}$$

where

$$\hat{P}_1^{cf}(z_{1-\alpha}) = -\hat{P}_1(z_{1-\alpha}); \quad (18)$$

$$\hat{P}_2^{cf}(z_{1-\alpha}) = \hat{P}_1(z_{1-\alpha})\hat{P}_1'(z_{1-\alpha}) - \frac{1}{2}z_{1-\alpha}(\hat{P}_1(z_{1-\alpha}))^2 - \hat{P}_2(z_{1-\alpha}); \quad (19)$$

$$\hat{P}_1(z_{1-\alpha}) = \hat{\kappa}_3(1 - z_{1-\alpha}^2)/6; \quad \hat{P}_1'(z_{1-\alpha}) = -\hat{\kappa}_3 z_{1-\alpha}/3; \quad (20)$$

$$\hat{P}_2(z_{1-\alpha}) = -z_{1-\alpha} \left\{ \frac{\hat{\kappa}_4(z_{1-\alpha}^2 - 3)}{24} + \frac{\hat{\kappa}_3^2(z_{1-\alpha}^4 - 10z_{1-\alpha}^2 + 15)}{72} \right\}; \quad (21)$$

$$\hat{\kappa}_3 = S_n^{-3}(\mathbf{X}_n)n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^3;$$

$$\hat{\kappa}_4 = S_n^{-4}(\mathbf{X}_n)n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^4 - 3.$$

Note that if we write $r_n := O_{a.s.}(n^{-3/2})$ for the rest term in (17), it then follows from Theorems 5.1 and 5.2 of Hall (1992) that $\limsup_{n \rightarrow \infty} n^{3/2}|r_n| \leq C$ a.s., for some finite constant $C > 0$. This, together with the Fatou-Lebesgue theorem imply that $E(r_n) = O(n^{-3/2})$, a fact that will be used throughout below.

From the results in Chapter 7 of Cramér (1945) it follows that

$$E(\hat{\kappa}_3) = \kappa_3 + O(n^{-1}), \quad E(\hat{\kappa}_4) = \kappa_4 + O(n^{-1}). \quad (22)$$

Applying an Edgeworth expansion (see, e.g., Hall, 1992) we deduce that

$$\begin{aligned} \varphi^0(\mathbf{X}_n) &\equiv P\left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n) | \mathbf{X}_n\right) \\ &= 1 - P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{\sigma} \leq \frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma} | \mathbf{X}_n\right) \\ &= 1 - \left\{ \Phi\left(\frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma}\right) + n^{-1/2} R_1\left(\frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma}\right) \right. \\ &\quad \left. + n^{-1} R_2\left(\frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma}\right) + O(n^{-3/2}) \right\}, \end{aligned} \quad (23)$$

where $O(n^{-3/2})$ is non-random and

$$R_j(x) = P_j(x)\phi(x); \quad (24)$$

$$P_1(x) = \kappa_3^0(1 - x^2)/6; \quad \kappa_3^0 = E((V_1^0 - \mu_0)^3)/\sigma^3; \quad (25)$$

$$\begin{aligned} P_2(x) &= -x \left\{ \frac{\kappa_4^0(x^2 - 3)}{24} + \frac{\kappa_3^{02}(x^4 - 10x^2 + 15)}{72} \right\}; \\ \kappa_4^0 &= \frac{E((V_1^0 - \mu_0)^4)}{\sigma^4} - 3. \end{aligned} \quad (26)$$

Note that $\kappa_3^0 = \kappa_3 = \mu_3/\sigma^3$ and $\kappa_4^0 = \kappa_4 = \mu_4/\sigma^4 - 3$ because of the assumption made on the distribution of V_1^0 . Let

$$\begin{aligned} c_{n,\alpha}(\mathbf{X}_n) &= z_{1-\alpha} + n^{-1/2} \hat{P}_1^{cf}(z_{1-\alpha}) + n^{-1} \hat{P}_2^{cf}(z_{1-\alpha}) \\ &\quad + O_{a.s.}(n^{-3/2}). \end{aligned} \quad (27)$$

Using (17), (27) and a Taylor series expansion, we find that

$$\begin{aligned}
& \Phi\left(\frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma}\right) \\
&= \Phi\left(\frac{S_n(\mathbf{X}_n)}{\sigma} \cdot c_{n,\alpha}(\mathbf{X}_n)\right) \\
&= 1 - \alpha + \phi(z_{1-\alpha}) \left\{ \hat{P}_1^{cf}(z_{1-\alpha})n^{-1/2} + \hat{P}_2^{cf}(z_{1-\alpha})n^{-1} \right\} \\
&\quad - z_{1-\alpha}\phi(z_{1-\alpha})\{\hat{P}_1^{cf}(z_{1-\alpha})\}^2n^{-1/2} \\
&\quad + (z_{1-\alpha}/\sigma)\phi(z_{1-\alpha})(S_n(\mathbf{X}_n) - \sigma) \\
&\quad - \sigma(z_{1-\alpha}/\sigma)^3\phi(z_{1-\alpha})(S_n(\mathbf{X}_n) - \sigma)^2/2 + O_{a.s.}(n^{-3/2}). \quad (28)
\end{aligned}$$

From (25), (27) and Taylor's theorem it easily follows that

$$\begin{aligned}
n^{-1/2}R_1\left(\frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma}\right) &= n^{-1/2}R_1\left(\frac{S_n(\mathbf{X}_n)}{\sigma} \cdot c_{n,\alpha}(\mathbf{X}_n)\right) \\
&= n^{-1/2}R_1(c_{n,\alpha}(\mathbf{X}_n)) + n^{-1/2}R'_1 \\
&\quad \times (c_{n,\alpha}(\mathbf{X}_n))c_{n,\alpha}(\mathbf{X}_n)(S_n(\mathbf{X}_n) - \sigma)/\sigma \\
&\quad + O_{a.s.}(n^{-3/2}) \\
&= n^{-1/2}P_1(z_{1-\alpha})\phi(z_{1-\alpha}) + n^{-1}\hat{P}_1^{cf} \times \\
&\quad (z_{1-\alpha})\phi(z_{1-\alpha}) \times \\
&\quad \{P'_1(z_{1-\alpha}) - z_{1-\alpha}P_1(z_{1-\alpha})\} \\
&\quad + O_{a.s.}(n^{-3/2}). \quad (29)
\end{aligned}$$

A first-order Taylor series expansion also yields

$$\begin{aligned}
n^{-1}R_2\left(\frac{C_{n,NP}^R(\alpha; \mathbf{X}_n)}{\sigma}\right) &= n^{-1}R_2\left(\frac{S_n(\mathbf{X}_n)}{\sigma} \cdot c_{n,\alpha}(\mathbf{X}_n)\right) \\
&= n^{-1}P_2(z_{1-\alpha})\phi(z_{1-\alpha}) + O_{a.s.}(n^{-3/2}). \quad (30)
\end{aligned}$$

From (18)–(30) we find, after tedious calculations, that

$$\begin{aligned}
\varphi^0(\mathbf{X}_n) &= \alpha - (z_{1-\alpha}/\sigma)\phi(z_{1-\alpha})(S_n(\mathbf{X}_n) - \sigma) \\
&\quad + \sigma(z_{1-\alpha}/\sigma)^3\phi(z_{1-\alpha})(S_n(\mathbf{X}_n) - \sigma)^2/2 \\
&\quad - n^{-1/2}(\hat{\kappa}_3 - \kappa_3)(z_{1-\alpha}^2 - 1)\phi(z_{1-\alpha})/6 \\
&\quad - n^{-1}(\hat{\kappa}_3 - \kappa_3)\hat{P}_1(z_{1-\alpha})\phi(z_{1-\alpha})z_{1-\alpha}(z_{1-\alpha}^2 - 3)/6 \\
&\quad + n^{-1}\{\hat{P}_2(z_{1-\alpha}) - P_2(z_{1-\alpha})\}\phi(z_{1-\alpha}) + O(n^{-3/2}) \\
&\quad + O_{a.s.}(n^{-3/2}). \tag{31}
\end{aligned}$$

Hence, from Lemma 1, (18)–(21), (22), (31) and the remark made above regarding $r_n := O_{a.s.}(n^{-3/2})$ we conclude that

$$\begin{aligned}
P\left(\sqrt{n}(\bar{V}_n^0 - \mu_0) \geq C_{n,NP}^R(\alpha; \mathbf{X}_n)\right) &= E_\mu(\varphi^0(\mathbf{X}_n)) \\
&= \alpha + (z_{1-\alpha}/\sigma)\phi(z_{1-\alpha})(\mu_4 + 3\sigma^4)/8n\sigma^3 \\
&\quad + \sigma(z_{1-\alpha}/\sigma)^3\phi(z_{1-\alpha})(\mu_4 - \sigma^4)/8n\sigma^2 \\
&\quad + O(n^{-3/2}) \\
&= \alpha + \frac{z_{1-\alpha}\phi(z_{1-\alpha})}{8n\sigma^4}\{z_{1-\alpha}^2(\mu_4 - \sigma^4) \\
&\quad + (\mu_4 + 3\sigma^4)\} + O(n^{-3/2}),
\end{aligned}$$

which completes the proof of Theorem 6.

A.4 Proof of Theorem 8

To prove Theorem 8, we shall make use of the following Lemma.

Lemma 2. Suppose that $E(|X_1|^6) < \infty$. Then, as $n \rightarrow \infty$,

$$E(\hat{\kappa}_3) = \kappa_3 + \frac{\mu_3(21\sigma^4 + 15\mu_4) - 12\sigma^2\mu_5}{8\sigma^7n} + O(n^{-2}).$$

Proof. Using Taylor's theorem we deduce that

$$\begin{aligned} E(\hat{\kappa}_3) &= \frac{E(m_3)}{(E(m_2))^{3/2}} + \frac{15}{8} \cdot \frac{E(m_3)}{(E(m_2))^{7/2}} \cdot \text{Var}(m_2) \\ &\quad - \frac{3}{2} \cdot \frac{1}{(E(m_2))^{5/2}} \cdot \text{Cov}(m_2, m_3) + O(n^{-2}), \end{aligned} \quad (32)$$

where $m_3 = n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^3$ and $m_2 = S_n^2(\mathbf{X}_n)$.

Note that from (27.4.5) and (27.5.1) of Cramér (1945) and Taylor's theorem we find that

$$\begin{aligned} \frac{E(m_3)}{(E(m_2))^{3/2}} &= \frac{1}{\sigma^3} \left(1 - \frac{1}{n}\right)^{-3/2} \left(1 - \frac{3}{n}\right) \mu_3 + O(n^{-2}) \\ &= \left(\frac{\mu_3}{\sigma^3}\right) \left(1 - \frac{3}{2n}\right) + O(n^{-2}). \end{aligned} \quad (33)$$

Equations (27.4.2) and (27.5.6) of Cramér (1945) also reveal that

$$\text{Var}(m_2) = \frac{\mu_4 - \sigma^4}{n} + O(n^{-2}), \quad (34)$$

and

$$\text{Cov}(m_2, m_3) = \frac{\mu_5 - 4\mu_3\sigma^2}{n} + O(n^{-2}). \quad (35)$$

The proof of the lemma now follows directly by substituting (33), (34) and (35) into (32).

The proof of the theorem will now be presented:

From Hall (1992) it follows that a Cornish-Fisher expansion for $C_{n,P}^R(\alpha; \mathbf{X}_n)$ is given by

$$\begin{aligned} C_{n,P}^R(\alpha; \mathbf{X}_n) &= z_{1-\alpha} + n^{-1/2} \hat{Q}_1^{cf}(z_{1-\alpha}) + n^{-1} \hat{Q}_2^{cf}(z_{1-\alpha}) \\ &\quad + n^{-3/2} \hat{Q}_3^{cf}(z_{1-\alpha}) + O_{a.s.}(n^{-2}), \end{aligned} \quad (36)$$

where

$$\hat{Q}_1^{cf}(z_{1-\alpha}) = -\hat{Q}_1(z_{1-\alpha}); \quad (37)$$

$$\hat{Q}_2^{cf}(z_{1-\alpha}) = \hat{Q}_1(z_{1-\alpha}) \hat{Q}_1'(z_{1-\alpha}) - \frac{1}{2} z_{1-\alpha} (\hat{Q}_1(z_{1-\alpha}))^2 - \hat{Q}_2(z_{1-\alpha}); \quad (38)$$

$$\hat{Q}_1(z_{1-\alpha}) = \frac{1}{6}\hat{\kappa}_3(1 + 2z_{1-\alpha}^2); \quad \hat{Q}'_1(z_{1-\alpha}) = \frac{2}{3}\hat{\kappa}_3 z_{1-\alpha}; \quad (39)$$

$$\begin{aligned} \hat{Q}_2(z_{1-\alpha}) = & z_{1-\alpha} \left\{ \frac{1}{12}\hat{\kappa}_4(z_{1-\alpha}^2 - 3) - \frac{1}{18}\hat{\kappa}_3^2(z_{1-\alpha}^4 + 2z_{1-\alpha}^2 - 3) \right. \\ & \left. - \frac{1}{4}(z_{1-\alpha}^2 + 3) \right\}. \end{aligned} \quad (40)$$

Furthermore, we require an expression for $\hat{Q}_3^{cf}(z_{1-\alpha})$. Using the relations given on page 69 of Hall (1992), laborious calculations yield the following expression (the calculations for deriving this are available on request from the authors):

$$\begin{aligned} \hat{Q}_3^{cf}(z_{1-\alpha}) = & -\frac{1}{3}(z_{1-\alpha}^2 - 1)\hat{Q}_1^3(z_{1-\alpha}) - \hat{Q}'_1(z_{1-\alpha})\hat{Q}_2^{cf}(z_{1-\alpha}) \\ & - \frac{1}{2}\hat{Q}_1''(z_{1-\alpha})\hat{Q}_1^2(z_{1-\alpha}) \\ & + \hat{Q}'_2(z_{1-\alpha})\hat{Q}_1(z_{1-\alpha}) + z_{1-\alpha}\hat{Q}_1^2(z_{1-\alpha})\hat{Q}'_1(z_{1-\alpha}) \\ & - z_{1-\alpha}\hat{Q}_1(z_{1-\alpha})\hat{Q}_2(z_{1-\alpha}) - \hat{Q}_3(z_{1-\alpha}). \end{aligned} \quad (41)$$

Applying an Edgeworth expansion (see, e.g., Hall, 1992) we derive that

$$\begin{aligned} \varphi^0(\mathbf{X}_n) \equiv & P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) | \mathbf{X}_n\right) \\ = & 1 - P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \leq C_{n,P}^R(\alpha; \mathbf{X}_n) | \mathbf{X}_n\right) \\ = & 1 - \{\Phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) \\ & + n^{-1/2}Q_1(C_{n,P}^R(\alpha; \mathbf{X}_n))\phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) \\ & + n^{-1}Q_2(C_{n,P}^R(\alpha; \mathbf{X}_n))\phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) \\ & + n^{-3/2}Q_3(C_{n,P}^R(\alpha; \mathbf{X}_n))\phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) + O(n^{-2})\}, \end{aligned} \quad (42)$$

where $O(n^{-2})$ is non-random,

$$Q_1(x) = \kappa_3(1 + 2x^2)/6, \quad (43)$$

and

$$Q_2(x) = x \left\{ \frac{1}{12} \kappa_4(x^2 - 3) - \frac{1}{18} \kappa_3^2(x^4 + 2x^2 - 3) - \frac{1}{4}(x^2 + 3) \right\}. \quad (44)$$

Using (36) and a Taylor series expansion, we find that

$$\begin{aligned} \Phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) &= 1 - \alpha + \phi(z_{1-\alpha}) \{ n^{-1/2} \hat{Q}_1^{cf}(z_{1-\alpha}) \\ &\quad + n^{-1} \hat{Q}_2^{cf}(z_{1-\alpha}) \\ &\quad + n^{-3/2} \hat{Q}_3^{cf}(z_{1-\alpha}) + O_{a.s.}(n^{-2}) \} \\ &\quad - z_{1-\alpha} \phi(z_{1-\alpha}) \{ n^{-1/2} \hat{Q}_1^{cf}(z_{1-\alpha}) \\ &\quad + n^{-1} \hat{Q}_2^{cf}(z_{1-\alpha}) \\ &\quad + O_{a.s.}(n^{-3/2}) \}^2 / 2. \end{aligned} \quad (45)$$

From (36) and Taylor's theorem it follows that

$$\begin{aligned} &n^{-1/2} Q_1(C_{n,P}^R(\alpha; \mathbf{X}_n)) \phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) \\ &= n^{-1/2} Q_1(z_{1-\alpha}) \phi(z_{1-\alpha}) + n^{-1/2} \{ Q_1'(z_{1-\alpha}) \\ &\quad - z_{1-\alpha} Q_1(z_{1-\alpha}) \} \phi(z_{1-\alpha}) \{ n^{-1/2} \hat{Q}_1^{cf}(z_{1-\alpha}) \\ &\quad + n^{-1} \hat{Q}_2^{cf}(z_{1-\alpha}) \} + O_{a.s.}(n^{-3/2}). \end{aligned} \quad (46)$$

A first-order Taylor series expansion also yields

$$\begin{aligned} &n^{-1} Q_2(C_{n,P}^R(\alpha; \mathbf{X}_n)) \phi(C_{n,P}^R(\alpha; \mathbf{X}_n)) \\ &= n^{-1} Q_2(z_{1-\alpha}) \phi(z_{1-\alpha}) + n^{-1} \{ Q_1'(z_{1-\alpha}) \\ &\quad - z_{1-\alpha} Q_1(z_{1-\alpha}) \} \phi(z_{1-\alpha}) \{ n^{-1/2} \hat{Q}_1^{cf}(z_{1-\alpha}) \\ &\quad + n^{-1} \hat{Q}_2^{cf}(z_{1-\alpha}) \} + O_{a.s.}(n^{-3/2}). \end{aligned} \quad (47)$$

From (36) – (47) we find, after laborious calculations, that

$$\begin{aligned}
\varphi^0(\mathbf{X}_n) = & \alpha + n^{-1/2}(1 + 2z_{1-\alpha}^2)\phi(z_{1-\alpha})(\hat{\kappa}_3 - \kappa_3)/6 \\
& - n^{-1}(\hat{\kappa}_3 - \kappa_3)\hat{Q}_1(z_{1-\alpha})\phi(z_{1-\alpha})z_{1-\alpha}(3 - 2z_{1-\alpha}^2)/6 \\
& + n^{-1}\{\hat{Q}_2(z_{1-\alpha}) - Q_2(z_{1-\alpha})\}\phi(z_{1-\alpha}) \\
& - n^{-3/2}\phi(z_{1-\alpha})\{Q_3(z_{1-\alpha}) - \hat{Q}_3(z_{1-\alpha})\} \\
& - n^{-3/2}\phi(z_{1-\alpha})\{-\hat{Q}_1^2(z_{1-\alpha})[\hat{Q}_1''(z_{1-\alpha}) - Q_1''(z_{1-\alpha})]/2 \\
& - \hat{Q}_2^{cf}(z_{1-\alpha})[(\hat{Q}_1'(z_{1-\alpha}) - Q_1'(z_{1-\alpha})) + z_{1-\alpha}(\hat{Q}_1(z_{1-\alpha}) \\
& - Q_1(z_{1-\alpha}))] - \hat{Q}_1(z_{1-\alpha})z_{1-\alpha}(\hat{Q}_2(z_{1-\alpha}) \\
& - Q_2(z_{1-\alpha})) + \hat{Q}_1^2(z_{1-\alpha})z_{1-\alpha}(\hat{Q}_1'(z_{1-\alpha}) - Q_1'(z_{1-\alpha})) \\
& + \hat{Q}_1(z_{1-\alpha})(\hat{Q}_2'(z_{1-\alpha}) - Q_2'(z_{1-\alpha})) - \hat{Q}_1^2(z_{1-\alpha})(z_{1-\alpha}^2 - 1) \\
& \times (\hat{Q}_1(z_{1-\alpha}) - Q_1(z_{1-\alpha}))/2\} + O_{a.s.}(n^{-2}). \tag{48}
\end{aligned}$$

Hence, from Lemma 2, (22), (37)–(41), (43), (44) and (48) we conclude that

$$\begin{aligned}
& P\left(\frac{\sqrt{n}(\bar{V}_n^0 - \mu_0)}{S_n(\mathbf{V}_n^0)} \geq C_{n,P}^R(\alpha; \mathbf{X}_n)\right) \\
& = E_\mu(\varphi^0(\mathbf{X}_n)) \\
& = \alpha + \frac{(1 + 2z_{1-\alpha}^2)\phi(z_{1-\alpha})\{\mu_3(21\sigma^4 + 15\mu_4) - 12\sigma^2\mu_5\}}{48\sigma^7n^{3/2}} \\
& \quad + O(n^{-2}),
\end{aligned}$$

which completes the proof of Theorem 8.

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